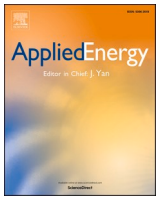
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Ecological network analysis and optimization of urban water metabolic system in context of energy nexus☆

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HIGHLIGHTS

- Network analysis and dual-objective optimization were combined for the urban water system.
- A water metabolic network (WMN) was proposed optimizing both resilience and energy consumption.
- 7.2 % of water resources and 5.5 % of energy consumption can be conserved by optimization of the WMN in Beijing.

ARTICLE INFO

Keywords:
Water metabolic network
Dual-objective optimization
Ecological network analysis
Energy consumption

ABSTRACT

The interconnectivity of energy and water resources is inherent within the urban water system. To meet the energy demands of the urban water system and optimize resource utilization, effective planning and management are essential. In this paper, a new framework incorporating network analysis and optimization model was developed to decrease energy consumption while enhancing water resource network resilience in Beijing, China. First, we established a water metabolic network (WMN) using ecological network analysis (ENA) to determine the water resource flow in each process. Then, the WMN performance was evaluated utilizing a set of indices derived from the ENA approach. Finally, an optimization model was set up with the objectives of minimizing energy consumption and maximizing network resilience, using non-dominated sorting genetic algorithm II (NSGA-II) to obtain the Pareto-optimal solutions. The results indicated that WMN was generally operated at a status of low efficiency, high redundancy and robustness. Totally 0.70 billion m^3 of water resources and 2.77 billion kWh of energy flow can be saved after the dual-objective optimization. The proposed framework may help optimize water metabolic network to decrease energy consumption while enhancing network resilience for a better urban water system planning, maintenance and management.

1. Introduction

The urban water systems are estimated to contribute 2–8 % of the global energy consumption. It is projected that with a potential 50 % increase in water usage by 2050, the energy demand for the urban water system will increase by 33 % [1,2]. In order to improve the energy management of the water system, it is crucial to identify the mutual dependencies between energy and water of urban system.

The urban water system includes water intake, water production, water delivery, water utilization, sewage treatment, collection and water reclamation. Energy plays a remarkable role in the functioning of the water system, e.g., pumping for water intake and delivery, electricity for water purification and sewage treatment processes, and residential heating and cooling, household cleaning, and washing [3].

Although there is a growing interest in comprehending the interdependencies between water and energy and the management implications for urban water systems, most studies have focused on single process rather than the entire system. A comparison of existing studies regarding the urban water-energy nexus can be seen in Table S1 in Supplementary Materials.

A metabolic network represents the flows of materials and energy within a complex system, reflecting interactions among its components, such as sectors or processes. Originally developed for biological systems,

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Nomenclature

Variable/Abbreviation	Description
WMN	Water metabolic network
ENA	Ecological network analysis
NSGA-II	Non-dominated sorting genetic algorithm II
LCA	Life cycle assessment
MFA	Material flow analysis
IOA	Input-output analysis
MRIO	Multi-regional input-output model
CGE	Computable general equilibrium
SD	System dynamics
f_{ij}	Water flow from node *j* to node *i*.
z_1	Input link from environment in the WMN
NE	Natural environment
DW	Domestic water sector
IW	Industrial water sector
AW	Agricultural water sector
WT	Wastewater treatment sector
RW	Reclaimed water sector
f_{12}	Direct discharge of sewage from the domestic sector to the natural environment
f_{13}	Direct discharge of sewage from the industrial sector to the natural environment
f_{14}	Direct discharge of sewage from the agricultural sector to the natural environment
f_{15}	Water discharged from the sewage treatment plant to the natural environment
f_{16}	Reclaimed water supplies ecological environment
f_{21}	Water extraction from the natural environment to the domestic sector
f_{26}	Reclaimed water used by the domestic sector
f_{31}	Water extraction from the natural environment to the industrial sector
f_{36}	Reclaimed water used by the industrial sector
f_{41}	Water extraction from the natural environment to the agricultural sector
f_{46}	Reclaimed water used by the agricultural sector
f_{52}	Domestic sewage quantity treated by sewage treatment plant
f_{53}	Industrial sewage quantity treated by sewage treatment plant
f_{54}	Agricultural sewage quantity treated by sewage treatment plant
f_{65}	Production of reclaimed water
AMI	Average mutual information
k	Scale coefficient
$\boldsymbol{F}_{..}$	Total water throughput includes the sum over all combinations of f_{ij}
$\boldsymbol{F}_{i.}$	Total water flow to node *i*
$\boldsymbol{F}_{.j}$	Total water flow from node *j*
H_c	Conditional entropy
$\boldsymbol{\alpha}$	Relative efficiency
R	Robustness
NCM	Network control measurement
N	Integral flow matrix
CD	Control difference matrix
SC	System control vector matrix
cd_{ij}	Control difference between sector *i* and *j*
sc_i	Control degree of sector *i* to the whole system
CR	Dimensionless control ratio matrix
cr_{ij}	Control influence of sector *j* on sector *i* via the integral system flow
F_1	Optimization function of energy consumption
F_2	Optimization function of network resilience
$\boldsymbol{\delta}$	Sewage treatment rate
W_1	Minimize restricted value of reclaimed water consumption
W_2	Total water use constraint value planned by the water management department minus the ecological water supplement of rivers and lakes
W_3	Minimum constraints for direct sewage discharge
l_d	Lower limit of water supply satisfaction rate for domestic sector
$\boldsymbol{l_i}$	Lower limit of water supply satisfaction rate for industry sector
$\boldsymbol{l_a}$	Lower limit of water supply satisfaction rate for agriculture sector
$\boldsymbol{u_d}$	Upper limit of water supply satisfaction rate for domestic sector
$\boldsymbol{u_i}$	Upper limit of water supply satisfaction rate for industry sector
$\boldsymbol{u_a}$	Upper limit of water supply satisfaction rate for agriculture sector
y	Value of sewage discharge coefficient

this concept has been adapted to urban systems to model the movement of resources such as water, energy, and materials across various sectors, including domestic, industrial, and agricultural, and to examine their interactions with the surrounding environment [4]. Some studies have further revealed the relationship between energy and water from the perspective of urban water metabolic network (WMN) [5], which provide insight into water efficiency, environmental ramifications, and impacts on external environment. Rather than focusing on individual processes, metabolic networks capture how various parts of a system work together, allowing for the identification of inefficiencies, redundancies, and opportunities for resource reuse. In order to specify the network attributes and systematic structural of urban water system, complex network such as ecological network analysis (ENA) has been widely applied [6]. ENA is considered as a favorable method to explore the relationship between the components of a system and reveal the integrity and complexity of the system [7]. Unlike traditional approaches that focus on individual components, ENA prioritizes the interactions between them, highlighting the dynamic attributes of the whole system.

Although many papers have described and accessed these relations, there is a gap on how to manage urban water system for energy saving and security of WMN. To balance the contradiction between water resource supply and demand, many researchers have utilized various methods to optimize the allocation of water system [8–12]. For example, the non-dominated sorting genetic algorithm II (NSGA-II) as an evolutionary solution algorithm has been designed to achieve near-optimal solutions for optimization models, which can serve as a useful tool to find optimal solutions of water-energy nexus. Moreover, increasing attention has been paid to involve the energy consumption and energy efficiency in the water system [13,14]. However, the current studies focus on a specific physical process of the water/energy system, while limited research has been conducted on the interaction between the attributes of sector nodes in the complex network formed by water and energy flows, as well as the optimal solutions and decision support for different policies and environmental changes.

Undoubtedly, energy saving remains a critical challenge in urban water resource management because energy is the necessary driving force for the utilization and consumption of water resources in all sectors

in cities. Meanwhile, the WMN is shaped by the flows and structure of various sectors associated with different attributes and characteristics of urban water system. The ability of water network responds and recovers after disturbances is thus crucial to ensure the resilience of urban water resource system. As such, the research motivation of this paper is to optimize water metabolic network to decrease energy consumption while enhancing network resilience for a better urban water system planning, maintenance and management. As the pressure on urban water systems increases, particularly in water-scarce regions, optimization of the water metabolic network becomes a critical tool for decision-makers. It can be used to determine which processes are most energy-intensive, where resources are being wasted, and how the system can be restructured to improve efficiency without compromising resilience. This framework is particularly valuable in developing synergistic strategies for water and energy management in cities facing both resource scarcity and growing populations.

This paper examines the implications of optimizing WMN to decrease energy consumption while enhancing water resource resilience. The research objectives that the study aims to address are as follows: (1) Quantification of the energy consumption within different processes of the WMN, encompassing water intake, production, delivery, utilization, sewage treatment, and recycling. (2) Introduction of a network perspective to elucidate the systemic attributes of the WMN, including dependence-control relationships, network efficiency, and redundancy. (3) Investigation of water flow optimization throughout the diverse processes of the water cycle, with the primary aim of mitigating energy consumption while concurrently bolstering the resilience of WMN.

We propose a framework combining ecological network analysis and dual-objective optimization to enhance urban water resource management. While traditional approaches to urban water management often focus on singular objectives, such as water conservation or energy reduction, this framework advances the field by simultaneously addressing energy efficiency and system resilience within an interconnected WMN. By utilizing the NSGA-II, the model identifies optimal solutions for both minimizing energy consumption and enhancing network robustness, thereby expanding the applicability of ENA to complex urban systems. This framework integrates energy saving considerations with the resilience of water metabolic networks, aiming to achieve synergistic savings in both water and energy while enhancing the resilience of urban WMN.

2. Literature review

2.1. Modelling of energy for urban water system

Over the past few decades, various methods have been employed to analyze the water-energy nexus across different contexts. Most researchers [15,16] used bottom-up methods to evaluate nexus, such as life cycle assessment (LCA) [17] and material flow analysis (MFA) [18]. For example, Pacetti et al. [19] investigated the trade-offs between water use and bioenergy production through the LCA methodologies. Tong et al. [20] proposed an MFA model that considered water use factors and system boundaries and quantified the water use and wastewater discharge inventories of different products. Such methods heavily depend on the accuracy of resource flow (i.e., water and energy) of the whole supply chain or specific process of products or substances, and thus, there are high requirements for available process databases [21].

Due to the limitations of data sets, particularly the challenges of obtaining process-level data, the top-down approaches are considered more convenient to calculate the water-energy nexus of industry or trade. The input-output analysis (IOA) can be applied to quantify both direct and indirect energy and water flows to calculate the quantity required to produce products and services, based on sectoral linkages in the trade process [22,23]. In order to quantify the water-energy nexus in different economic sectors in the indirectly process of trade, such as the energy embodied in economic activities in water systems, multi-regional input–output (MRIO) model was adopted [24]. Wang and Chen [25] inventoried the energy-related water and water-related energy systematically with the MRIO method, showing that sectoral water-related energy flows within the Beijing Tianjin Hebei region account for 87 % of the total, of which Beijing, Tianjin, and Hebei account for 13 %, 24 %, 50 %, respectively. The top-down approach is based on inter-sectoral linkages through input-output table data, and may oversimplify the complexity of the water-energy nexus at the local level and cannot capture the specific characteristics of individual processes or industries [26]. Consequently, simulation models, such as the computable general equilibrium (CGE) model and system dynamics (SD) model, have been used to describe the energy-water nexus in different processes of a single system [27,28]. For example, Zhou et al. [29] built a multi-sectoral dynamic CGE model with an energy tax module to study the impact on energy and water resources. Cai et al. [30] coupled the SD model, orthogonal design, and copula analysis to assess coupled risks of energy-water nexus, illustrated the shortage risks of water and energy resources, and identified the controlled amount of water and energy demands.

Recently, integrated models have been applied to quantify the relationship in the complex systems, which are a combination of different single models with optimized, quantitative and scenarios functions [31]. Zhang and Vesselinov [32] developed a synergistic approach that utilizes systems analysis and mathematical modelling to enhance decision makers' comprehension of the complex interrelationships and trade-offs between energy and water resources. Chen and Chen [33] proposed a system-based framework to synthesize the interwoven connections between energy consumption and water use in a city and assessed the intensities of embodied energy and water triggered by urban consumption. However, the research system boundaries in the aforementioned methods have been constrained by the availability of a database, which makes it difficult to meet the characteristics of complex multi-process and multi-agent urban water systems.

2.2. ENA for urban water network

ENA, first proposed by Hannon [34], is a system-oriented method designed to examine the flow of materials through an ecosystem. ENA aims to analyze the indirect impacts on the system by considering the interactions among all sectors and the overall structure of the system [35]. Research on network structure characteristics, such as efficiency, redundancy and robustness of water resources network based on ENA, is of great significance for improving water use efficiency and water system stability [36,37]. Bodini and Bondavalli [38] studied the flow exchange of urban water supply system among different active departments, established the flow network of an urban water supply system, and analyzed its sustainability. By utilizing the tool from ENA, control analysis and utility analysis between different water sectors can be explored to clarify the relationship between different water use departments [39]. Fang and Chen [40] used the ENA method to simulate a virtual water network, identifying the water interdependency among different sectors as well as the network efficiency and stability in a socio-economic system. Most recently, the use of ENA has allowed researchers to identify key species and interactions in water-energy nexus networks, and to assess the potential impacts of changes in the availability or use of water or energy resources on ecosystem structure and function. Wang et al. [41] identified the control and dependence relationships by combining direct energy/water networks and hybrid energy and water networks. A comprehensive assessment for energy-water metabolic network based on ENA was established by Zhang et al. [42] to clarify the embodied linkages and the complicated system interactions in the energy-water nexus network.

3. Methodology

3.1. Water metabolic network model

ENA simulates the interactions among organisms, describing the structure of material and energy flow between different compartments within a system. Its fundamental units consist of compartments and pathways, in which, compartments refer to specific functional units within the system, while pathways denote the channels through which material and energy are transferred between compartments. Here WMN refers to the interconnected relationship of water flow, distribution, and usage within urban area. A WMN encompasses various components such as water sources, distribution systems, consumption patterns, treatment facilities, and environmental impacts. From an ENA perspective, we constructed a WMN model to demonstrate the structure and function of urban water system among varies sectors, in which, node represents specific sector and directional line represents the water flow [43,44].

The construction of the WMN involves three steps: (I) System boundary: Defining the system boundary according to the research purpose. A time-series inventory of city-related water resource flows, inputs, or emissions within the system boundary is required to quantify the variation of direct metabolic process. (II) Flow relationship: Identifying the flow path of water resources and the transfer relationship of water resources between different sectors. The functional shift of the water metabolic networks is assessed at sectoral and process levels. (III) Network construction: Compiling WMN for various processes in different sectors with adjacency matrix according to the flow relationship.

As shown in Fig. 1, the WMN contains 6 sectors (NE, DW, IW, AW, WT, RW) and 15 flow links. The sectors serve as specific functional units that require initial identification. To delineate the physical water flows associated with natural water resources utilized by urban sectors, the compartment of natural environment (NE) is employed to supply fresh water to fulfill the requirements of various sectors, encompassing domestic (DW), industrial (IW), and agricultural (AW) compartments. DW encompasses water necessary for daily life and production activities, including households, offices, and commercial establishments. IW encompasses water resources utilized in industrial production processes, spanning activities such as manufacturing, processing, cooling, and cleaning. AW pertains to water utilized for irrigation and livestock. The wastewater treatment (WT) compartment indicates that wastewater discharged by water-using sectors can be collected by urban wastewater treatment facilities [36,45,46]. Reclaimed water (RW) denotes the portion that can be reused by urban water systems following wastewater treatment.

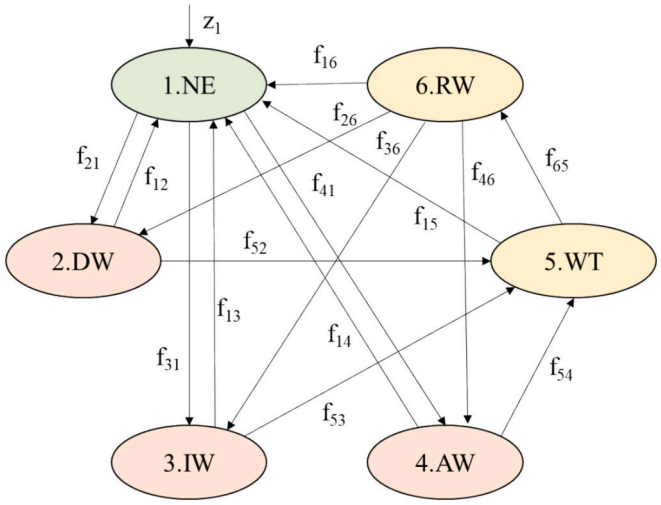


Fig. 1. Scheme of WMN model for Beijing.
Note: Nature environment (NE), Domestic water (DW), Industrial water (IW), Agricultural water (AW), Wastewater treatment (WT), Reclaimed water (RW). f_{12}: Direct discharge of sewage from the domestic sector to the natural environment; f_{13}: Direct discharge of sewage from the industrial sector to the natural environment; f_{14}: Direct discharge of sewage from the agricultural sector to the natural environment; f_{15}: Water discharged from the sewage treatment plant to the natural environment; f_{16}: Reclaimed water supplies ecological environment; f_{21}: Water extraction from the natural environment to the domestic sector; f_{26}: Reclaimed water used by the domestic sector; f_{31}: Water extraction from the natural environment to the industrial sector; f_{36}: Reclaimed water used by the industrial sector; f_{41}: Water extraction from the natural environment to the agricultural sector; f_{46}: Reclaimed water used by the agricultural sector; f_{52}: Domestic sewage quantity treated by sewage treatment plant; f_{53}: Industrial sewage quantity treated by sewage treatment plant; f_{54}: Agricultural sewage quantity treated by sewage treatment plant; f_{65}: Production of reclaimed water.

The f_{ij} between each link represents the water flow from node j to node i, and the arrow from j to i. The input link from the environment of the system is represented by z_i, and the arrow points to the component.

3.2. Information-based ENA

3.2.1. Efficiency calculation

The network efficiency in the WMN refers to the effective utilization of water and energy in the network. Average mutual information (*AMI*) quantifies the amount of flow diversity encumbered by structural constraints [47,48] and the correlation between components and reflect the development of positive feedback in the system [49]. Thus, it can be employed to gauge the degree of orderly and coherent connection among flows within the network, and serve as a crucial index for efficiency analysis, offering insights into the overall organization and effectiveness of resource flows within the system. A WMN with a high *AMI* value suggests that the network flows are more concentrated with fewer dispersion pathways, indicating that it is better organized and has higher efficiency [50]. *AMI* is calculated as follows:

$$AMI = k\sum_{i=1}^{n+2}\sum_{j=0}^{n}\frac{f_{ij}}{F_{..}}log_2\frac{f_{ij}F_{..}}{F_{i.}F_{j}} \tag{1}$$

where *AMI* (i.e., $K = 1$) represents the most primitive efficiency result of the network; k is the scale coefficient. f_{ij} represents the water flow from node j to node i; $F_{..}$ represents the total water throughput includes the sum over all combinations of f_{ij}; $F_{i.}$ represents the total water flow to node i; F_j represents the total water flow from node j.

3.2.2. Redundancy calculation

Network redundancy refers to the network's capacity to sustain its function and performance despite external interferences or internal failures. Conditional entropy (H_c) quantifies the level of "choice" or residual diversity/freedom associated with both the inputs and outputs of an average compartment within the network [45,51,52]. Therefore, H_c is utilized to measure the disorder and freedom retained by the network and is chosen as the metric for redundancy analysis. H_c is calculated as follows:

$$H_c = -k\sum_{j=0}^{n}\frac{f_{ij}}{F_{..}}log_2\left(\frac{f_{ij}^{\,2}}{F_{i.}F_{j}}\right) \tag{2}$$

A higher H_c value indicates that the water flow structure of the WMN is less organized, i.e., more alternate links exist in the network to contain unidentified flow, corresponding to higher redundancy. While certain degree of redundancy can enhance the ability of the WMN structure to mitigate various risks, excessive redundancy can diminish the efficiency of the system, which is not conducive to its sustainable development. It is important to strike a balance between redundancy and efficiency to ensure the WMN structure is able to function optimally and maintain its sustainability in the long run.

3.2.3. Robustness calculation

Robustness (R) refers to the ability of the WMN system to maintain stability when the system suffers risks and faults. Robustness can indicate both the stability of the system and the development potential of the system. Therefore, it can be measured by the balance between the two indicators of comprehensive efficiency and redundancy [40]. A relative measurement of the organized power flowing within the WMN, namely, the proportion of development capacity accounted for by the order degree (α, i.e., relative efficiency) is denoted by the following Eq. 3.

$$\alpha = \frac{\mathrm{AMI}}{\mathrm{AMI} + \mathrm{H_c}} \tag{3}$$

Then, R, which indicates both the stability of the system and development potential of the system, can be measured by [53]:

$$R = -\alpha log_2(\alpha) \tag{4}$$

3.2.4. Network control measurement (NCM)

NCM is employed to assess the influence of one sector over another by examining pairwise interactions, which can enable the identification of the impact of water resource flow control at a particular node on other nodes. This method specifically focuses on the identification of the influence of department nodes on the network [54]. Typically, the integral flow matrix is defined as N and N':

$$\mathrm{N} = \left[\mathrm{n_{ij}}\right] = (\mathrm{I} - \mathrm{G})^{-1} \tag{5}$$

$$\mathrm{N'} = \left[\mathrm{n_{ij}^{-1}}\right] = (\mathrm{I} - \mathrm{G'})^{-1} \tag{6}$$

where $G = \left[g_{ij}\right]$, $g_{ij} = f_{ij}/F_j$, $G' = \left[g_{ij}{}'\right]$, and $g_{ij}{}' = f_{ij}/F_i$.

The control difference matrix (CD) and the system control vector matrix (SC) can be defined to encourage the explanation of pairwise individual comparisons of the WMN flow and transfer.

$$\mathrm{CD} = \left[\mathrm{cd_{ij}}\right] = \left[\mathrm{n_{ij}} - \mathrm{n_{ji}}\right] \tag{7}$$

$$\mathrm{SC} = \left[\mathrm{sc_i}\right] = \left[\sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{cd_{ij}}\right] \tag{8}$$

where cd_{ij} is the control difference between sector i and j. sc_i is the control degree of sector i to the whole system. If sc_i value is positive, it indicates that sector i has a control effect on the system, while if not, sector i is controlled by the system.

The dimensionless control ratio matrix (CR) expresses the comparison between the two sectors:

$$CR = \left[cr_{ij}\right] = \begin{cases} n_{ij} - n_{ji} > 0, cr_{ij} = \left[\dfrac{n_{ij} - n_{ji}}{max(n_{ij}, n_{ji})}\right] \\ n_{ij} - n_{ji} \le 0, cr_{ij} = 0 \end{cases} \tag{9}$$

where cr_{ij} is the control influence of sector j on sector i via the integral system flow.

3.3. Dual-objective optimization model

3.3.1. Optimization objective

A dual-objective optimization model aimed at minimizing energy consumption and maximizing network robustness is proposed to optimize the performance of WMN. The resilience of WMN refers to the characteristics of the system that respond to specific disturbances and its ability to maintain a variety of original functions, or the overall operational efficiency of the network when the system is faced with the impact of changes in the internal structure or external environment. This model can help to achieve energy efficiency and enhance the network resilience, thereby contributing to its overall sustainability and long-term viability [55]. The objective function can be expressed as:

$$F_1 = min\{E_t\} \tag{10}$$

$$F_2 = max\{R\} \tag{11}$$

where E_t is the total energy consumption, kWh; The detailed energy consumption calculation process can be seen in Note S1 of Supplementary Materials. R is the robustness of WMN.

3.3.2. Optimization constraints

WMN follows the physical constraints, such as the sewage treatment rate constraint, reclaimed water development constraint, amount of available water resources constraint, direct discharge quantity constraint, water supply guarantee rate constraint, emission coefficient constraint and variable nonnegative constraint. The mathematical formulations of these constraints are given as follows:

$$\frac{\mathrm{f_{15}} + \mathrm{f_{65}}}{\left(\mathrm{f_{52}} + \mathrm{f_{53}} + \mathrm{f_{54}}\right)} \ge \delta \tag{12}$$

where δ is the sewage treatment rate.

$$\mathrm{f_{16}} + \mathrm{f_{26}} + \mathrm{f_{36}} + \mathrm{f_{46}} \ge \mathrm{W_1} \tag{13}$$

where W_1 is the minimize restricted value of reclaimed water consumption, m^3.

$$\mathrm{f_{21}} + \mathrm{f_{31}} + \mathrm{f_{41}} + \mathrm{f_{26}} + \mathrm{f_{36}} + \mathrm{f_{46}} \le \mathrm{W_2} \tag{14}$$

where W_2 is the total water use constraint value planned by the water management department minus the ecological water supplement of rivers and lakes, m^3.

$$\mathrm{f_{12}} + \mathrm{f_{13}} + \mathrm{f_{14}} \le \mathrm{W_3} \tag{15}$$

where W_3 is the minimum constraints for direct sewage discharge, m^3.

$$\mathrm{l_d}\left(\mathrm{f_{21}} + \mathrm{f_{26}}\right) \le \mathrm{f_{21}} + \mathrm{f_{26}} \le \mathrm{u_d}\left(\mathrm{f_{21}} + \mathrm{f_{26}}\right) \tag{16}$$

$$\mathrm{l_i}\left(\mathrm{f_{31}} + \mathrm{f_{36}}\right) \le \mathrm{f_{31}} + \mathrm{f_{36}} \le \mathrm{u_i}\left(\mathrm{f_{31}} + \mathrm{f_{36}}\right) \tag{17}$$

$$\mathrm{l_a}\left(\mathrm{f_{41}} + \mathrm{f_{46}}\right) \le \mathrm{f_{41}} + \mathrm{f_{46}} \le \mathrm{u_a}\left(\mathrm{f_{41}} + \mathrm{f_{46}}\right) \tag{18}$$

where l_d, l_i, l_a represent the lower limit of water supply satisfaction rate for domestic, industry, and agriculture sectors, respectively; u_d,u_i, u_a represent the upper limit of water supply satisfaction rate for domestic, industry and agriculture sectors, respectively.

$$\mathrm{f_{12}} + \mathrm{f_{52}} \ge \mathrm{y} \times \left(\mathrm{f_{21}} + \mathrm{f_{26}}\right) \tag{19}$$

$$\mathrm{f_{13}} + \mathrm{f_{53}} \ge \mathrm{y} \times \left(\mathrm{f_{31}} + \mathrm{f_{36}}\right) \tag{20}$$

$$\mathrm{f_{14}} + \mathrm{f_{54}} \ge \mathrm{y} \times \left(\mathrm{f_{41}} + \mathrm{f_{46}}\right) \tag{21}$$

where y is the value of sewage discharge coefficient.

$$\mathrm{f_{ij}} \ge 0 \tag{22}$$

where f_{ij} is the water flow from sector j to sector i.

$$\mathrm{f_{out}}/\mathrm{f_{in}} \ge 0.9 \tag{23}$$

where f_{in} and f_{out} are the total inflows and outflows for one sector, respectively.

3.3.3. Optimization algorithm

NSGA-II employs the fundamental principles of genetic algorithms, simulating natural selection and genetic mechanisms to generate and maintain a population of solutions that evolve and improve throughout the search process [56]. This evolutionary process enables NSGA-II to

efficiently explore a high-quality solution space, aiding better understanding of the intricate dynamics within water-energy coupling relationships [57,58]. The constructed WMN contains 6 sectors and 15 decision variables, with 11 constraint conditions and 2 optimization objectives. For optimization processes with few objectives but large computational load, NSGA-II demonstrates its advantages with strong convergence and diversity, producing a range of non-dominated solutions covering the entire Pareto front. This diverse set of solutions provides decision-makers with multiple options, facilitating a better understanding of trade-offs and interconnections among different research objectives [59,60]. Therefore, we choose the NSGA-II algorithm to solve the dual-objective problem of WMN.

In order to evaluate the fitness functions based on the objective functions, the initialized population size, pareto fraction, and maximum generation pf iteration of iterations of the algorithm are presented for the WMN simulation. If all constraints are met, solutions are then ranked using a fast non-dominated approach to identify different Pareto rank levels. In cases where the fitness values of two competing designs are equal, the crowding distance of each individual is calculated based on the average Euclidean distance to measure proximity between an individual and its neighbors. Generation involves binary tournament selection based on rank and crowding distance to select superior solutions from the existing population. The selected population generates offspring through crossover and mutation operators. This process continues for subsequent generations until the stopping criterion is reached. The try and error method was adopted to determine the NSGA-II parameters. We selected a total of 81 parameter simulation schemes for papulation size (100,200,300), maximum generation (100, 200, 300), crossover probability (0.8, 0.9, 1.0) and mutation probability (0.03, 0.05, 0.1) to analyze the uncertainty of the model. The 95 % confidence interval of the optimized energy consumption is 478.37×10^8 kWh-540.07×10^8 kWh. The 95 % confidence interval of the optimized network robustness is 0.398–0.459. In the parameter setting of the NSGA-II, the population size is set as 200, the crossover probability 0.9, and the maximum generations of iterations 300. Further details on the NSGA-II can refer to Deb et al. [61], and Liu et al. [62].

4. Study area and data

Beijing, as the capital of China, faces significant challenges with regards to water shortages. In the period from 2011 to 2020, the city experienced an average precipitation of only 569 mm, with per capita water resources of 135m^3, 1/16 of the country [63]. Despite these challenges, Beijing has implemented impressive water conservation measures. Total water consumption in the city remained below 4.1 billion m^3 between 2011 and 2020, while water consumption per ten thousand yuan of GDP decreased by 52.4 % from 15m^3 to 7m^3 [64]. Since 2004, Beijing has incorporated reclaimed water into its unified water resource allocation strategy, with reclaimed water contributing to approximately one-third of the city's total water consumption, and aimed to increase the utilization rate of reclaimed water to over 35 % by 2025 [65]. The inter-basin water transfer projects, such as the South-to-North Water Diversion Project (SNWD), primarily involve diverting water resources from the Yangtze River, the Yellow River, and other southern water sources to the northern region. This initiative further alleviates the increasing demand of water resources for urban and industrial development, especially the over-exploitation of groundwater in Beijing.

In 2018, Beijing ‘s total energy consumption was 72.70 million tons of standard coal, with coal, oil and natural gas respectively accounting for 2.77 %, 34.14 % and 34.17 %, and non-fossil energy accounting for 7.8 % of the total. The production and supply of water alone accounted for the consumption of 649,400 tons of standard coal, representing a significant portion of the energy consumption within the water industry [66].

The water resources data used in the literature are from the Beijing Water Statistical Yearbook 2018, Beijing Water Resources Bulletin 2018, Urban Water Supply Statistical Yearbook 2018, Urban and Rural Construction Statistical Yearbook 2018, and China Water Resources Bulletin 2018. The pollution emission data are from the Beijing Environmental Statistics Annual Report 2018. The socio-economic development data pertaining to Beijing are sourced from two key publications: the Beijing Statistical Yearbook 2018 and the China Statistical Yearbook 2018. The data regarding future development planning pertain to Beijing City Master Plan (2016–2035), the “14th Five-Year” urban sewage treatment and resource utilization development plan 2021 as well as the “14th Five-Year Plan for Beijing's national economic and social development” and “2035 Outline of long-term goals”. These plans offer critical insights into the strategic vision and objectives for urban development of Beijing, providing a comprehensive framework for sustainable growth and prosperity in the city. The detailed parameters and values of energy consumption and optimization constrain in Beijing are shown in Tables S2 and S3 in Supplementary Materials.

5. Results and discussion

5.1. Water flow and energy consumption in WMN

The total water resource flow of the WMN in Beijing is 9.67 billion m^3. It can be primarily divided into four categories: surface water, groundwater, reclaimed water, and inter-basin water transfer, accounting for 7.11 %, 38.10 %, 33.12 %, and 21.67 %, respectively (Table 1). Due to the scarcity of surface water and the need to protect ecological water use, the amount of surface water utilization in Beijing is strictly limited. Of all the water use sectors, domestic water consumption in Beijing accounts for the highest proportion, mounting to 66.42 % of total water consumption, far exceeding industrial and agricultural water consumption. This water consumption structure is related to the dominance of tertiary industries, such as high-tech and science and education industries, in Beijing's industrial structure. Meanwhile, the large number of permanent residents in Beijing implies that ensuring domestic water consumption is the primary task in optimizing water resource allocation (See Fig. 2).

In the context of energy consumption intensity within water use sector, the domestic water sector exhibits the highest intensity at 28.90 kWh/m^3, followed by the industrial water sector. Among all processes, the surface water intake and water supply processes exhibit relatively low energy consumption intensity, approximately one thousandth of the domestic water sector. The energy consumption intensity of SNWD is 9.8 times that of groundwater, 50 times that of surface water, and 2.8 times that of reclaimed water. Consequently, in terms of the water-energy nexus, it is more economical, environmentally friendly, and sustainable to vigorously promote the development of reclaimed water resources in Beijing in the future.

Table 1
Energy and water flows of WMN in Beijing.

Num	Process		Energy intensity (kWh/m^3)	Water resources ($10^8 m^3$)	Energy consumption (10^8kWh)
1	Water intake	Surface	0.03	2.31	0.07
2		Groundwater	0.16	12.38	1.94
3		SNWD	1.54	7.04	10.84
4	Water supply		0.03	22.57	0.63
5	Water use	Domestic water	28.90	14.99	433.21
6		Industrial water	11.00	3.40	37.40
7		Agricultural water		4.18	
8	Wastewater treatment		0.53	19.03	10.05
9	Reclaimed water		0.53	10.76	5.74
Total				96.66	499.88

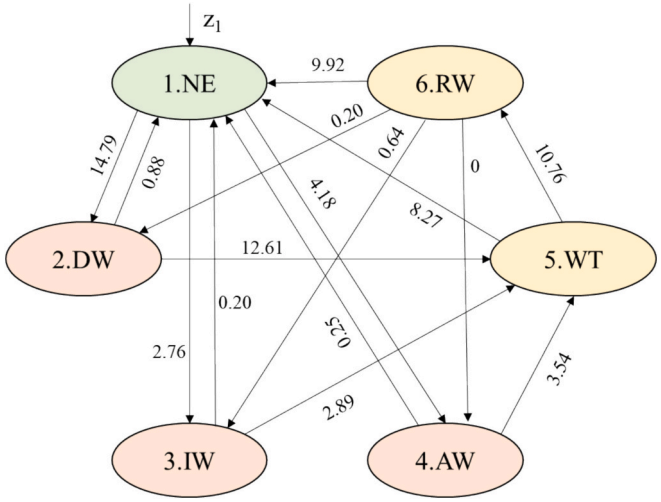


Fig. 2. Water flow network of Beijing before optimization.
Note: Nature environment (NE), Domestic water (DW), Industrial water (IW), Agricultural water (AW), Wastewater treatment (WT), Reclaimed water (RW).

The total energy consumption of the WMN in Beijing is 49.99 billion kWh, of which the water use process has the highest energy consumption, accounting for 94 % of the total energy consumption, with 43.32 billion kWh. DW energy consumption is the highest in the water use process, accounting for 86.66 % of the total energy consumption. The water intake process had the second-highest energy consumption, accounting for about 2.57 % of the total energy consumption, with SNWD being the dominant process, accounting for 2.17 % of the total energy consumption and 84.36 % of the energy consumption of the water intake process. The energy consumption of water supply is the lowest, with 0.63 billion kWh. The processes are ranked from highest to lowest energy consumption as follows: water use>water intake> drainage and sewage treatment> reclaimed water>water supply.

5.2. Information-based ENA for WMN

The findings reveal that the *AMI* of the Beijing WMN system is 0.44, the H_c is 14.51, and the R is 0.15. The outcomes demonstrate that the water resource utilization redundancy of the WMN system is high. However, the system efficiency is low, which hinders the system's ability to achieve a low level of robustness. This results in the network's insufficient capacity to mitigate risks, ultimately leading to low overall resilience of the water resources network.

NE, DW, and WT departments exhibit a robust control relationship, with NE and WT exerting considerable control over the WMN in Beijing. Meanwhile, IW, AW, and RW sections demonstrate pronounced dependence, with IW and AW displaying the most notable dependency. Among the controlling sectors, the NE sector is the dominant controller. The control degrees of NE over IW, NE over AW, WT over IW, and WT over AW are all above 80 %. However, the control degree of AW over IW is below 20 % (See Table 2).

5.3. Pareto frontier solutions

Fig. 3 presents the Pareto frontier of EC and NR, which contains a set of non-dominated solutions containing 150 solutions. It illustrates the trade-off relationship between EC and NR, where an increase in the resilience of the water resources network will bring about the increase of EC. Increasing network resilience typically requires a series of infrastructure and operational adjustments designed to ensure the system remains functional during disruptions, such as extreme weather events, system failures, or demand fluctuations. To achieve this, redundancy is often incorporated into the network, which involves adding backup pathways, storage facilities, and treatment capacities. For example, redundant water distribution pipelines and additional storage reservoirs ensure that water can be rerouted in case of pipeline failures or supply interruptions. Similarly, backup treatment plants or secondary purification processes ensure water quality is maintained even if primary systems are compromised. While these measures enhance the system's ability to recover from disturbances, they also demand additional energy to operate.

The trade-off between energy consumption and network resilience carries significant implications for urban water management, offering critical insights for decision-makers as they navigate the balance between operational efficiency and system robustness. The Pareto frontier analysis provides a spectrum of non-dominated solutions, enabling policymakers to select configurations that align with local priorities and constraints. In regions where water supply disruptions can lead to significant social and economic impacts, such as in agricultural areas or industrial hubs, investing in resilience may be the most cost-effective choice in the long run. On the other hand, in areas with less vulnerability to disruptions, decision-makers may opt to prioritize energy efficiency by reducing redundancy or focusing on optimizing existing

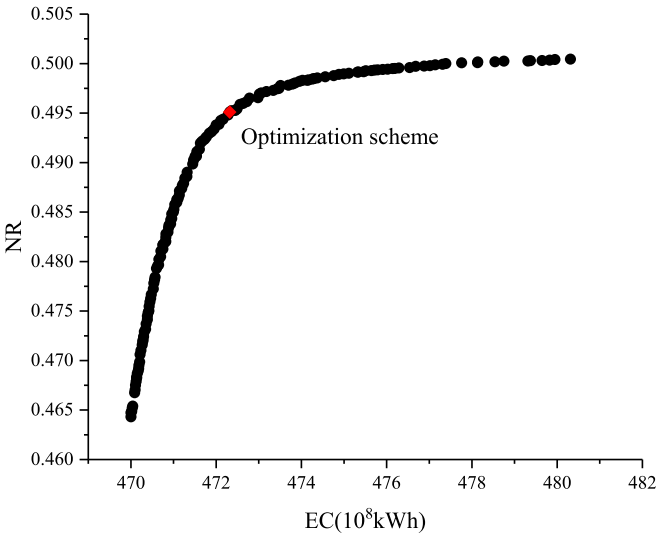


Fig. 3. Pareto frontier for WMN optimization with two objectives

Table 2
Control ratio matrix and control/dependence degree of WMN in Beijing before optimization.

Control ratio matrix	NE	DW	IW	AW	WT	RW
NE	0	0.31	0.84	0.81	0.03	0.45
DW	0	0	0.77	0.72	0	0.20
IW	0	0	0	0	0	0
AW	0	0	0.17	0	0	0
WT	0	0.29	0.84	0.80	0	0.43
RW	0	0	0.71	0.65	0	0
Control/dependence intensity	1.98E16	4.71E15	−2.13E16	−1.97E16	1.86E16	−2.14E15

Note: Nature environment (NE), Domestic water (DW), Industrial water (IW), Agricultural water (AW), Wastewater treatment (WT), Reclaimed water (RW).

infrastructure. The trade-off offers a framework for cities to highlight the importance of planning for dual-purpose infrastructure that address both energy and water needs, ultimately fostering resilient and sustainable urban water systems.

The weights of the two optimization objectives, i.e., energy consumption and network resilience, were set equally. Also, the WMN of Beijing in 2018 was selected to emphasize the energy nexus for comparative analysis.

5.4. Optimization results analysis

(1) Water resource and energy flow network.

Figs. 4 and 5 show the optimized water resources for different processes and WMN changes. The overall optimized water flux is 8.97 billion m^3, which is reduced by 0.70 billion m^3 compared with pre-optimization (9.67 billion m^3). The WMN matrices of before and after optimization are shown in Tables S4 and S5 in Supplementary Materials. The analysis of each process in the WMN reveals that the difference in water resource flux across the optimization primarily occurs in the sewage collection and treatment process. Specifically, the amount of sewage collection and treatment before optimization is 1.90 billion m^3, while optimized result reduces this value by 332.00 million m^3. Therefore, it calls for an improvement in the utilization rate of water resources, particularly in the domestic water sector, to reduce the generation of sewage.

(2) Energy consumption of WMN.

Fig. 6 shows a total energy consumption reduction of 2.77 billion kWh of WMN, which accounts for 5.54 % decrease compared to the original state. Furthermore, the DW sector has the highest potential for energy saving, exhibiting a 5 % reduction in energy consumption after optimization, followed by IW and SNWD.

(3) Network robustness analysis.

The optimized result of *R* exhibits significant improvement with 2.3 times increase. The value of *AMI* after optimization also has a remarkable increase of 81.82 %. However, the value of *Hc* is decreased, which results in less redundancy of WMN (See Table 3). As can be seen in Fig. 7, the optimized results are located on the left side of the curve, indicating a higher robustness value compared to the original state.

When one part of the WMN system is changed, the entire system consequently gets affected owing to the interconnectedness in the context of water-energy nexus [67]. The failure of critical infrastructures within the energy, water, and allied sectors results in the imbalance of links and paths, thus impacting the whole system. Insufficient development of water infrastructures, including purification, storage, transportation technologies, sanitation, and maintenance of water supply,

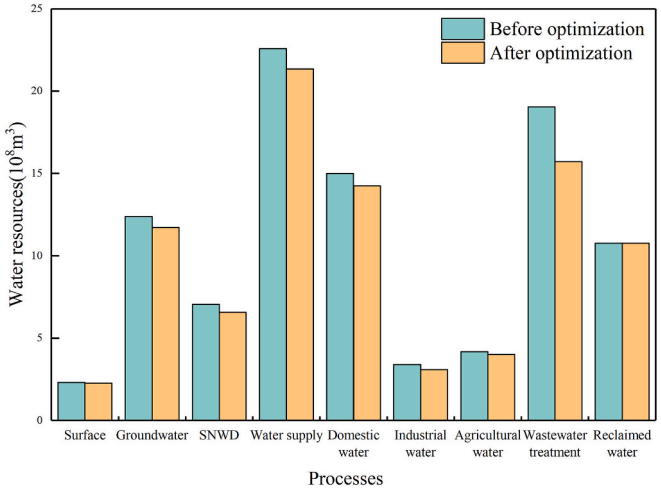


Fig. 4. Process-based water flow before and after optimization in Beijing.

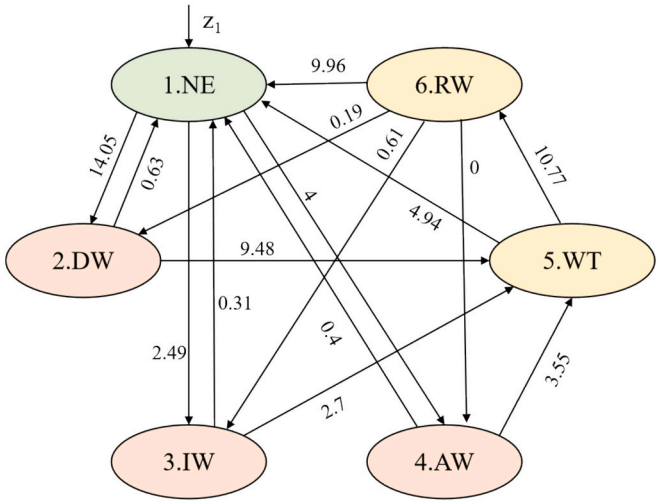


Fig. 5. Water flow network of Beijing after optimization.
Note: Nature environment (NE), Domestic water (DW), Industrial water (IW), Agricultural water (AW), Wastewater treatment (WT), Reclaimed water (RW).

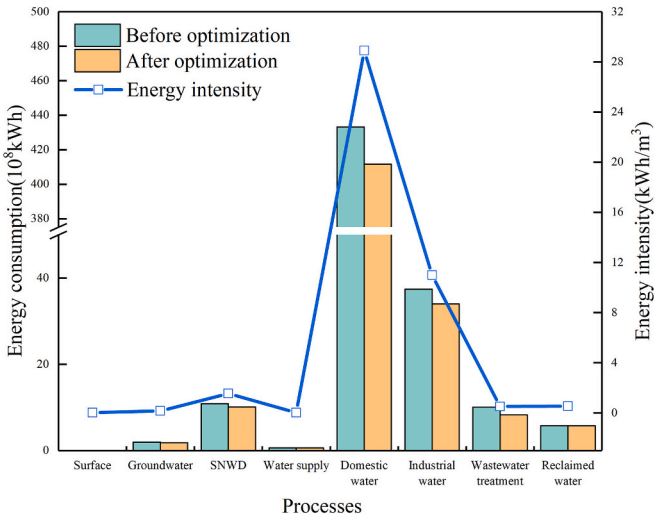


Fig. 6. Process-based energy intensity and energy consumption of Beijing.

Table 3
Indicators of information-based ENA.

Results	*AMI*	*Hc*	*α*	*R*
Before optimization	0.44	14.51	0.03	0.15
After optimization	0.80	2.49	0.24	0.50

poses significant risks to water security. Similarly, energy security is susceptible to uncontrollable exploration, mining, transportation, conversion, storage and utilization [68]. The escalation of extreme weather events and the failure to adequately mitigate or adapt to climate change contribute to a scarcity of available water resources in natural environments or an increased demand for energy extraction, further exacerbating the risk to water security and generating an imbalance between energy requirements and supplies [69].

(4) Network control analysis.

The system control/dependence degrees of each sector are shown in Fig. 8 based on the results of the SC matrix. The control degree of each sector is spread in a large scale after optimization. The RW sector changes from a dependent sector to a controlling one after optimization,

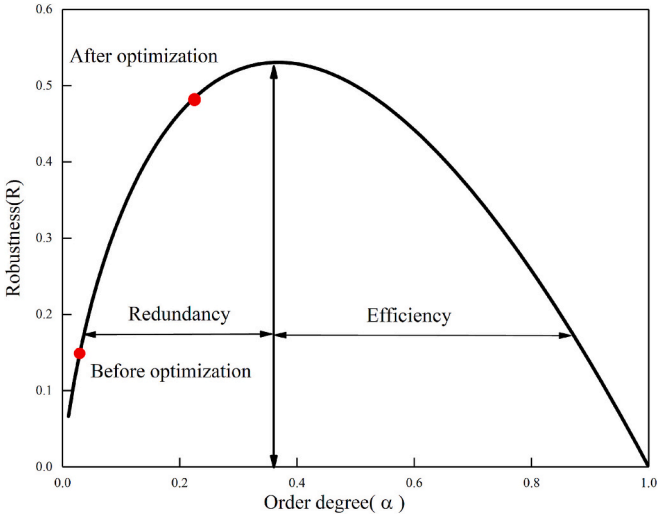
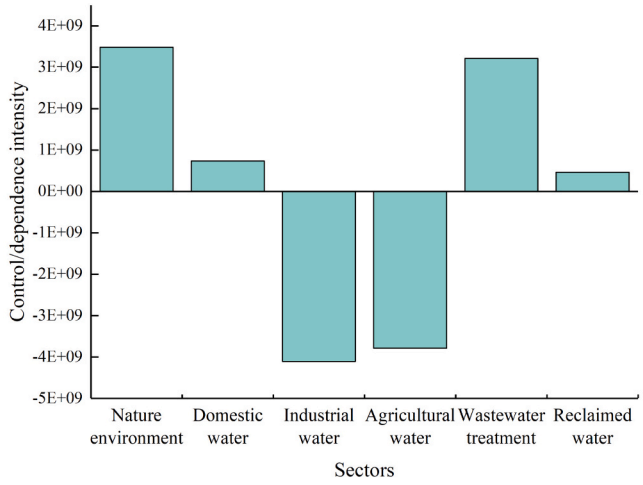
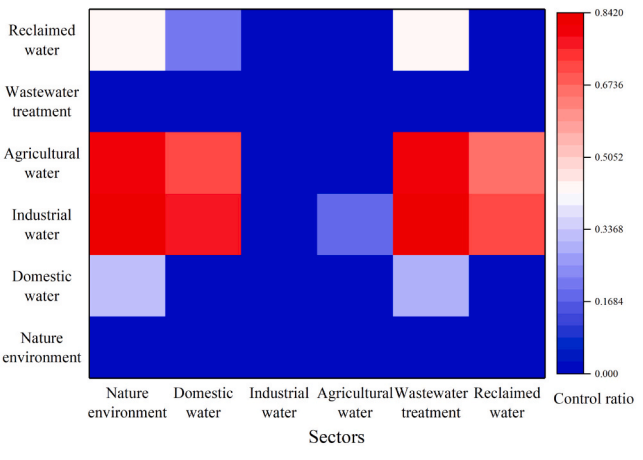


Fig. 7. Relationship between α and *R* in the WMN.



(a)



(b)

Fig. 8. Network control analysis of WMN in Beijing after optimization. (a) Control/dependence degree. (b) Control ratio (CR) matrix

and the control-dependence relationships of the other sectors do not change. IW and AW are the main dependent sectors, and NE and WT are the main controlling sectors.

Fig. 8 (b) illustrates that NE and WT are the two main dominators of the WMN. IW and AW rely on other sector water suppliers. The control degrees of NE on IW and WT on IW are above 80 %. For the optimization scheme, the control degrees of NE on AW and WT on AW are below 80 %. The control degrees of RW on DW have increased. The dependence of IW and AW on other sectors are nearly 90 %, indicating a heavy reliance on water imports from outside.

Except for the dominant sectors, the stronger and weaker pathways can be identified. The control degree higher than 50 % represents strong linkage between pair-wise sectors including DW-IW, DW-AW, RW-IW and RW-AW, while those lower than 50 % embrace NE-DW, NE-WT, NE-RW, DW-RW, AW-IW, WT-DW and WT-RW.

5.5. Discussion

The optimization of WMN offers a potential solution for advancing water conservation and energy efficiency in Beijing. According to Beijing's Water Resources Security Plan (2020–2035), the total urban water consumption is limited to 4.3 billion m^3. Meanwhile, in line with Beijing's Energy Development Plan for the 14th Five-Year Plan period, efforts have been directed towards controlling total energy consumption at 67.62 million tons of standard coal.

By optimizing the water metabolic network, the synergistic effect of reducing water flow and saving energy consumption can be achieved. The most notable optimization processes are the reduction of effluent discharges from the domestic water sector and the wastewater treatment plants. In terms of energy saving, the energy consumption of SNWD, domestic water and sewage treatment are the main energy-intensive processes. Particularly, the SNWD has been put into operation through a 9-stage pumping station, which annually consumes 0.39 billion kWh and needs improving energy-saving techniques and optimal scheduling of cascade pumping stations along the water transmission line.

It is also important to recognize that there exists a delicate balance between network resilience and the utilization of reclaimed water. Therefore, pursuing external water diversion projects, such as SNWD, represents a crucial external strategy for addressing the challenges related to water resources saving and network resilience improvement in Beijing.

1) Economic impact of optimizing the water metabolic network.

The optimization of the WMN offers significant potential for economic benefits, primarily through reduced energy consumption and enhanced water resource management. We calculated the economic impact of optimizing various processes, as detailed in Table S8 in Supplementary Materials. The optimization resulted in a decrease in energy consumption by approximately 2.77 billion kWh in Beijing, which translates to a monetary saving of 1.49 billion yuan. The optimization model enhanced water resource management, enabling a more efficient allocation of water resources and yielding savings of 0.70 billion m^3. This translates an additional economic benefit of 2.23 billion yuan.

The indirect economic benefits extend beyond direct cost savings. By enhancing the resilience of the WMN, cities can mitigate the economic impacts associated with water shortages, supply disruptions, and infrastructure failures. For example, maintaining reliable water services can support economic stability and growth, particularly in sectors reliant on consistent water availability, such as agriculture and manufacturing. Quantifying these broader economic impacts could further enhance the justification for investments in optimized water management strategies, emphasizing their importance not only for sustainability but also for long-term economic viability.

2) Risks of development of reclaimed water in future.

Reclaimed water, even after treatment, may still contain various types of chemical and biological contaminants that can persist in the environment or bioaccumulate. Pathogens, including bacteria, viruses,

protozoa, and antibiotic-resistant genes, are common in treated wastewater and may not be fully removed by conventional treatment processes, posing potential risks for infection in humans and animals. Heavy metals, such as cadmium, lead, and mercury, are often present in trace amounts in reclaimed water and can accumulate in soils and crops, leading to long-term health risks, including kidney damage and developmental delays. Pharmaceuticals and personal care products (PPCPs), such as hormones, antibiotics, and other bioactive compounds, are typically not completely eliminated during treatment and may disrupt endocrine function in both humans and wildlife [70–73]. In addition, organic contaminants, including pesticides, herbicides, and industrial chemicals like bisphenol A, further complicate the management of reclaimed water [74]. These compounds have the potential to public health and ecosystems, highlighting the challenge of ensuring safe reclaimed water use.

The presence of contaminants in reclaimed water raises significant concerns for public health, especially among vulnerable populations such as children, the elderly, and immunocompromised individuals. Health risks from reclaimed water exposure can occur through direct and indirect pathways. Direct exposure, such as skin contact or inhalation of aerosols during irrigation with reclaimed water, increases the risk of infection from pathogens. Indirect exposure through the food chain is also a concern; crops irrigated with reclaimed water may absorb heavy metals and pharmaceuticals, which then enter the human food chain. Long-term consumption of produce irrigated with contaminated water could lead to bioaccumulation, resulting in chronic exposure to low levels of pharmaceuticals or trace heavy metals, with potential impacts on human health, including hormonal disruptions and increased cancer risk.

In addition to human health risks, the use of reclaimed water poses potential ecological impacts. In agricultural systems, plants and soil are susceptible to these contaminants when reclaimed water is used for irrigation. Long-term irrigation with reclaimed water can lead to the accumulation of heavy metals and microcontaminants in soil, posing persistent risks [75,76]. In river ecosystems, the introduction of reclaimed water as a replenishment source poses threats to the health of aquatic vegetation, groundwater and aquifer. Studies have shown that the replenishment of rivers with reclaimed water can induce toxicity in zebrafish, affecting various physiological systems including the nervous, hepatic, endothelial-vascular, and myeloid systems [77,78].

3) Urban planning and policy implications for sustainable water management.

The study illustrates how Beijing can improve efficiency and enhance resilience without large-scale infrastructure investments, by reducing water consumption by 0.70 billion m^3 and decreasing energy consumption by 2.77 billion kWh. Urban planners can leverage this approach to address future water and energy demands, especially in regions where urban expansion and population growth are expected to outpace resource availability. For example, the results highlight the crucial role of reclaimed water in reducing both water and energy demands. Integrating reclaimed water into a comprehensive water management strategy, as demonstrated in Beijing, could be applied to other water-stressed cities, potentially decreasing dependence on long-distance water transfers and reducing over-extraction from natural sources. Policymakers can use these insights to craft policies that encourage the reuse of water and the reduction of energy-intensive processes, particularly in water-stressed areas. For instance, the reduction in energy consumption through better management of the SNWD in Beijing highlights how improving operational efficiency can lead to significant savings. Moreover, the framework provides policymakers with a tool to assess the trade-offs between energy consumption and network resilience. Cities that are particularly vulnerable to water supply disruptions due to climate change or aging infrastructure can benefit from the resilience-enhancing features of the WMN, which ensure that water systems can withstand and recover from disturbances more effectively. Policies that promote the resilient design of water networks, such as incorporating redundancy into critical infrastructure and encouraging cross-sectoral resource flows (e.g., wastewater reuse), can play a key role in sustainable urban water management.

4) Comparison with other results.

In comparative analysis of constructed WMN structure, we developed 6 sectors and 15 flow links in water metabolic network. Cui et al. [45] conducted further refinement of the industrial sector by constructing 8 sectors and 41 links, which emphasizes the impact of the primary, secondary, and tertiary industries. Zhang et al. [46] defined 18 metabolic links that reflect the flows among these 6 sectors, which focuses on the effect of rainwater collection on the network. The water metabolic network constructed by different studies varies mainly due to differing research priorities. We constructed 6 sectors and 15 links, which explores the role of reclaimed water as a separate component to the WMN in Beijing.

The results in this paper were compared with previous studies. Huang et al. [79] analyzed the water-energy relationship of the water supply system in Beijing, which indicated that the energy intensity of surface water intake ranged from 0.018 kWh/m^3 to 0.18 kWh/m^3. They found that the energy intensity of reclaimed water treatment was 0.82 kWh/m^3, and the energy intensity of sewage treatment was 0.32 kWh/m^3. The energy intensity for surface water intake calculated in our study was 0.029 kWh/m^3, the energy intensity of reclaimed water treatment was 0.53 kWh/m^3. These results are different from those of this paper due to the mismatch between statistical time scales and data sources.

The optimization model predicted a reduction of approximately 0.70 billion m^3 in water usage and 2.77 billion kWh in energy consumption for the Beijing case study. To validate these projections, we analyzed actual water consumption and energy usage data from the Beijing Water Statistical Yearbook 2023 and the Beijing Energy Consumption Report 2023 for the same period. Historical data indicated that prior water conservation initiatives achieved savings of approximately 0.10 billion m^3 annually. Additionally, the Beijing Municipal Development and Reform Commission reported that comprehensive energy conservation measures have led to energy savings of about 14.717 billion kWh. Considering that the water resources system accounts for roughly 5 % of the city's total energy consumption, it can be concluded that the energy savings attributable to Beijing's water system are approximately 735 million kWh. These real-world data closely align with the model's predictions, demonstrating its capability to produce realistic and reliable outcomes.

In addition, we explored the relationship between energy consumption and network resilience, incorporated energy consumption and network resilience into urban water resources management, and explored the optimization of urban water resources network structure for water conservation and energy saving as well as network resilience enhancement.

5) Future work.

The constructed WMN contains the processes of water intake, water supply, water use, drainage and sewage treatment, and reclaimed water, and the water resource processes in other cities are basically the same as those mentioned above, manifesting the potential to be applied to other cities. Also, we consider the inter-basin water transfer project of SNWD, which involves 40 cities in 7 provinces along the water transfer route, and the framework can be better used in the study of energy-water nexus of the above cities.

Considering that urban resources encompass water, energy, land, mineral resources and ecosystems, although the current study focuses on the energy and water, the proposed framework can be transplanted to the coordinative management of other resources like land, food, and mineral resources.

6. Conclusions and limitations

6.1. Conclusions

This research provides a novel contribution to urban resource management by demonstrating a framework that integrates ecological network analysis with dual-objective optimization to address the energy and resilience needs of urban water systems. The study advances current knowledge by illustrating how a WMN can be optimized not only for resource efficiency but also for resilience against disturbances. By applying the NSGA-II algorithm, the framework generates a spectrum of solutions along a Pareto frontier, enabling urban planners to make informed decisions based on trade-offs between energy use and system robustness. This approach provides a scalable model for cities facing water scarcity and energy challenges, offering a practical methodology that could guide future developments in energy-water nexus strategies and sustainable urban water management. Through this research, the integration of energy consumption and resilience into urban water optimization models marks a significant step forward in creating adaptable, resource-efficient cities.

The total water flow and energy consumption of WMN of Beijing 2018 were 9.67 billion m^3 and 49.99 billion kWh, respectively. After dual-objective optimization of WMN, the water flow for WMN decreased by 0.70 billion m^3, of which WT sector exhibited the largest reduction. The energy consumption for urban water system was reduced by 2.77 billion kWh, of which DW contributed most.

After optimization of WMN, the *AMI* value representing efficiency increased from 0.44 to 0.80, demonstrating a significant increment of 81.82 %. The *Hc* value representing redundancy, however, decreased by 82.84 %. The R value representing robustness was improved by 2.3 times. Generally, the WMN of Beijing was operated at a status of low efficiency, high redundancy and robustness.

Network control analysis of WMN showed that the NE and WT sectors were the primary controlling sectors, while the IW and AW sectors exhibited the strongest dependence. After optimization, although the control/dependence relationship of each sector remains unchanged, the degree of control/dependence increased. The control degrees of NE on IW and WT on IW were above 80 %, while the dependence degrees of IW and AW on the other sectors were nearly 90 %.

6.2. Limitations

The energy consumption associated with river and lake ecological water replenishment is not considered in this paper. Such replenishment is achieved through two methods: reducing the amount of water taken from surface water resources and implementing artificial water replenishment measures. The artificial water replenishment facilities involve gate pump scheduling after water diversion. Since the energy consumption of gate-pump switching scheduling process represents a small proportion of the overall energy consumption of the urban water system, we choose to ignore this aspect of energy consumption.

Our constructed WMN and optimization model is static and at one year-scale. Actually, the water resources keep cycling and never stop in the urban water metabolic system. The cyclic process is not accounted for in the WMN, and also not considered in the optimization model. The primary challenge for dynamic modelling lies in acquiring real-time data on water resource flow and the data fusion at various scales. The framework should be further improved to provide a more precise simulation of the complexity of the real urban water system.

CRediT authorship contribution statement

Jiaxiang Zou: Writing – original draft, Visualization, Investigation, Conceptualization. **Bin Chen:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition. **Saige Wang:** Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2025.125700.

Data availability

The authors do not have permission to share data.

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